

EthiViz User Guide

Welcome to EthiViz, a comprehensive platform for analyzing cultural bias in both text and image data through multiple ethical traditions. This guide will help you navigate the platform's features and functionalities.

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Getting Started

Installation

1. Ensure you have Python 3.8 or higher installed.
2. Install the required dependencies:

```
pip install -r requirements.txt
```

For optimal performance with image analysis, we recommend installing the optional dependencies:

```
pip install opencv-python tensorflow
```

Basic Usage

To run a basic analysis with default settings:

```
python run.py --text-data path/to/textdata.csv --image-dir path/to/image/
```

Command Line Usage

EthiViz provides a flexible command-line interface with numerous options:

```
python run.py [options]
```

Common Options

Option	Description	Default
<code>--text-data</code>	Path to text dataset (CSV, JSON, or Excel)	None
<code>--image-dir</code>	Path to directory containing images	None
<code>--traditions</code>	Comma-separated list of ethical traditions	western,ubuntu,confucian,islamic
<code>--feature-level</code>	Feature extraction level for image analysis	medium
<code>--batch-size</code>	Batch size for processing images	32
<code>--output-dir</code>	Directory to save analysis results	./results
<code>--port</code>	Port for the visualization dashboard	8050
<code>--no-dashboard</code>	Skip launching the dashboard	False

Examples

Analyze text data only:

```
python run.py --text-data data/articles.csv
```

Analyze images only with basic feature extraction:

```
python run.py --image-dir data/marketing_images --feature-level basic
```

Analyze both with specific ethical traditions:

```
python run.py --text-data data/social_media.csv --image-dir data/profile_
```

Save results to a custom directory:

```
python run.py --text-data data/survey.csv --output-dir ./my_analysis_resu
```

Run analysis without launching the dashboard:

```
python run.py --text-data data/comments.csv --no-dashboard
```

Python API Usage

For programmatic integration or custom workflows, you can use the Python API:

```
from text_analyzer import TextAnalyzer
from image_analyzer import ImageAnalyzer
from visualization import create_dashboard

# Text analysis
text_analyzer = TextAnalyzer(traditions=['western', 'ubuntu', 'confucian']
text_results = text_analyzer.analyze('path/to/data.csv')

# Image analysis
image_analyzer = ImageAnalyzer(feature_level='medium', batch_size=32)
image_paths = ['path/to/image1.jpg', 'path/to/image2.jpg']
image_results = image_analyzer.analyze_batch(image_paths)

# Create dashboard
create_dashboard(text_results=text_results, image_results=image_results,
```

Customizing Text Analysis

```
analyzer = TextAnalyzer(
    traditions=['western', 'ubuntu'], # Only use Western and Ubuntu trac
    nlp_model='en_core_web_md',      # Specify SpaCy model
    max_tokens=10000                 # Limit token count for performar
)
```

Customizing Image Analysis

```
analyzer = ImageAnalyzer(  
    feature_level='advanced',          # Use deep learning features  
    traditions=['confucian', 'islamic'], # Only use Confucian and Islamic  
    batch_size=64,                    # Process 64 images at once  
    detect_demographics=True,          # Enable demographic detection  
    detect_cultural_elements=True      # Enable cultural element detecti  
)  
  
# Analyze a single image  
result = analyzer.analyze_image('path/to/image.jpg')  
  
# Access specific metrics  
skin_tone_diversity = result['skin_tone_diversity']  
confucian_ethics_score = result['confucian_ethics_score']
```

Using the Interactive Dashboard

The interactive dashboard provides comprehensive visualizations of analysis results.

Dashboard Layout

The dashboard is organized into three main tabs:

1. **Text Analysis:** Visualizations for text data
2. **Image Analysis:** Visualizations for image data
3. **Combined Analysis:** Comparative visualizations between text and image results

Navigating Visualizations

- **Hover** over data points for detailed information
- Use the **toolbar** in the upper right of each chart for interactions:
 - Pan: Move around the visualization
 - Box Select: Select data points
 - Zoom: Zoom in/out
 - Download: Save the chart as PNG
 - Reset: Return to the original view

Filtering Data

Some visualizations offer filtering options:

- Use dropdown menus to select specific ethical traditions
- Adjust sliders to filter data points by score ranges
- Click on legend items to toggle specific data series

Text Analysis

Supported Data Formats

- CSV files
- JSON files
- Excel spreadsheets (.xlsx, .xls)

Key Metrics

- **Ethics Scores:** Assessment of content through different ethical traditions
- **Bias Score:** Overall measure of potential bias
- **Diversity Index:** Measure of representation diversity
- **Sentiment Analysis:** Positive/negative/neutral sentiment detection
- **Cultural Context Sensitivity:** How content relates to different cultural contexts

Image Analysis

Supported Image Formats

- JPEG (.jpg, .jpeg)
- PNG (.png)
- GIF (.gif)
- BMP (.bmp)

Feature Extraction Levels

1. **Basic** (Uses Pillow)
 - Color histograms
 - Basic shape analysis
 - Simple skin tone detection
2. **Medium** (Requires OpenCV)

- HOG features
- More accurate skin tone detection
- Improved shape and texture analysis

3. **Advanced** (Requires TensorFlow)

- Deep learning features
- Enhanced demographic detection
- Cultural element recognition

Key Metrics

- **Skin Tone Distribution:** Approximation of Fitzpatrick scale representation
- **Gender Distribution:** Estimated gender representation
- **Age Distribution:** Estimated age group representation
- **Cultural Element Detection:** Identification of cultural symbols, clothing, etc.
- **Ethics Scores:** Assessment through different ethical traditions
- **Diversity Index:** Shannon entropy-based measure of representation diversity

Interpreting Results

Ethics Scores

Ethics scores range from 0 to 10, where:

- **0-3:** Potentially problematic from this ethical tradition's perspective
- **4-6:** Neutral or mixed assessment
- **7-10:** Aligned with this ethical tradition's principles

Diversity Index

The diversity index ranges from 0 to 1, where:

- **0:** No diversity (complete homogeneity)
- **0.5:** Moderate diversity
- **1:** Maximum diversity (perfect representation balance)

Bias Score

Bias scores range from 0 to 10, where:

- **0-3:** Low potential for bias

- **4-6:** Moderate potential for bias
- **7-10:** High potential for bias

Ethical Traditions Metrics

Western Metrics

- **Fairness:** Equal treatment and opportunity across groups
- **Autonomy:** Respecting individual choice and freedom
- **Non-maleficence:** Avoiding harm to individuals
- **Justice:** Distribution of benefits and burdens

Ubuntu Metrics

- **Community Harmony:** Impact on group relationships and cohesion
- **Communal Benefit:** Collective positive outcomes for all groups
- **Relational Fairness:** How relationships between groups are affected
- **Shared Dignity:** Recognition of common humanity across groups

Confucian Metrics

- **Role Appropriateness:** Alignment with social roles and expectations
- **Social Harmony:** Maintenance of balanced social relationships
- **Reciprocity:** Balance of giving and receiving across groups
- **Ritual Propriety:** Adherence to cultural appropriateness

Islamic Metrics

- **Adl (Justice):** Fairness in distribution and treatment
- **Ihsan (Excellence):** Going beyond bare minimum in ethical treatment
- **Karama (Dignity):** Preserving human dignity across all groups
- **Maslaha (Public Interest):** Benefit to the collective community

Advanced Configurations

Performance Optimization

For large datasets:

- Use `--batch-size` to adjust processing chunk size
- Select appropriate `--feature-level` based on computational resources

- Use `--no-dashboard` to skip the interactive visualization
- Consider using a subset of ethical traditions with `--traditions`

Processing Multiple Files

You can process multiple files by using wildcards in bash:

```
# Create a list of image files
find data/marketing_images -name "*.jpg" > image_list.txt

# Process images in batches
python run.py --image-dir data/marketing_images --batch-size 50
```

Customizing Visualization Appearance

You can customize dashboard appearance by modifying the visualization.py file:

```
# In visualization.py
# Change color scheme
color_scheme = {
    'western': '#003366',
    'ubuntu': '#006633',
    'confucian': '#660033',
    'islamic': '#333366'
}

# Apply to visualizations
fig.update_traces(marker_color=color_scheme[tradition])
```

Troubleshooting

Common Issues

Issue: Dashboard shows "No results available"

Solution: Ensure your data paths are correct and the files contain valid data

Issue: Image analysis is slow

Solution: Use a smaller batch size or lower feature extraction level

Issue: Missing dependencies error

Solution: Run `pip install -r requirements.txt` to install required packages

Issue: Memory errors with large datasets

Solution: Process data in smaller chunks or reduce feature extraction complexity

Getting Help

If you encounter problems not covered in this guide:

- Check the project README for additional information
- Look for error messages in the console output
- Review the logs in the output directory

Visualization Types

The EthiViz dashboard includes several key visualization types:

Text Analysis Visualizations

- **Ethics Scores Box Plots:** Compare scores across different ethical traditions
- **Bias Distribution Histogram:** Distribution of bias scores across the dataset
- **Diversity vs. Bias Scatter Plot:** Relationship between diversity and bias metrics

Image Analysis Visualizations

- **Skin Tone Distribution Chart:** Shows representation across skin tone categories
- **Gender Distribution Chart:** Display gender representation in images
- **Diversity Index Bar Chart:** Diversity scores by image
- **Ethics Scores by Tradition:** Compare how images score across ethical traditions
- **Image Gallery:** Display of analyzed images with their key metrics

Combined Analysis Visualizations

- **Text vs. Image Bias Comparison:** Box plots comparing bias scores
- **Diversity Index Comparison:** Compares diversity between text and image data
- **Ethics Score Comparison:** Comparison of ethical perspectives across data types

Ethical Traditions

EthiViz analyzes data through multiple ethical traditions:

Western Ethics

Based on principles like autonomy, beneficence, non-maleficence, and justice. Western ethics prioritizes individual rights, fairness, and equal treatment.

Ubuntu Ethics

Stems from the African philosophy of "I am because we are." Ubuntu ethics emphasizes communal relationships, shared humanity, and collective well-being over individual interests.

Confucian Ethics

Focuses on proper relationships, social harmony, and virtues. Confucian ethics emphasizes role fulfillment, hierarchical relationships, and the importance of societal order.

Islamic Ethics

Centers on justice, dignity, and community welfare. Islamic ethics balances individual rights with societal obligations and emphasizes the inherent dignity of all people.

Using the Streamlit Interface

EthiViz now includes a modern, intuitive Streamlit interface that makes it easy to analyze data without writing code.

Starting the Application

To launch the Streamlit interface:

```
streamlit run app.py
```

This will open a web browser with the EthiViz application.

Interface Layout

The interface consists of:

1. **Sidebar:** Contains all input controls and analysis settings

- Data source selection
- Ethical tradition selection
- Advanced options
- Run button

2. **Main Area:** Displays analysis results in a tabbed interface

- Text Analysis tab
- Image Analysis tab
- Combined Analysis tab

Workflow

1. **Select Data Sources:**

- For text data: Upload a CSV/Excel file or use sample data
- For image data: Upload images or use sample images

2. **Configure Analysis:**

- Select which ethical traditions to include
- Adjust advanced options if needed (feature level, batch size, etc.)

3. **Run Analysis:**

- Click the "Run Analysis" button
- A progress bar will show analysis status

4. **Explore Results:**

- Navigate between tabs to view different aspects of the analysis
- Interact with visualizations (hover, zoom, download)
- View raw data tables

Data Upload Guidelines

- **Text Data:** CSV or Excel files with text content
- **Images:** JPG, JPEG, PNG, or GIF files
- **File Size:** Keep individual files under 200MB for best performance